

RELI-RAG

Network & Security Benchmark Reference

A practice document for testing retrieval, evidence citation, answer faithfulness, relevance, and insufficient-evidence behavior.

Document type	Benchmark reference material
Intended use	Local UI and retrieval evaluation practice
Content	40 distinct technical topics
Important note	Synthetic educational content; not an operational security standard.

How to use this document

Ask questions whose answers are explicitly present in the reference. Check whether the UI can show source names and page references. Include questions that combine two concepts and questions that cannot be answered from this document. For unsupported questions, the expected behavior is to explain that the available evidence is insufficient.

01. Network Fundamentals

A computer network connects devices so they can exchange data and share resources. Networks are commonly described by scope: PAN, LAN, MAN, and WAN. A LAN typically serves a room, building, or campus; a WAN connects geographically separated networks. Network design balances performance, availability, security, cost, and manageability.

Key points

- A computer network connects devices so they can exchange data and share resources.
- Networks are commonly described by scope: PAN, LAN, MAN, and WAN.
- A LAN typically serves a room, building, or campus; a WAN connects geographically separated networks.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

02. OSI Reference Model

The OSI model describes communication in seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. Ethernet framing is associated with the Data Link layer, IP routing with the Network layer, and TCP/UDP with the Transport layer. The model is a troubleshooting aid, not a requirement that every protocol map perfectly to one layer.

Key points

- The OSI model describes communication in seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application.
- Ethernet framing is associated with the Data Link layer, IP routing with the Network layer, and TCP/UDP with the Transport layer.
- The model is a troubleshooting aid, not a requirement that every protocol map perfectly to one layer.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

03. TCP and UDP

TCP provides a reliable, ordered byte stream using connection establishment, sequence numbers, acknowledgements, retransmission, and flow control. UDP sends independent datagrams without built-in delivery guarantees or ordering. DNS queries often use UDP, while applications may use TCP for larger responses or specific operational needs. The application determines which transport is appropriate.

Key points

- TCP provides a reliable, ordered byte stream using connection establishment, sequence numbers, acknowledgements, retransmission, and flow control.
- UDP sends independent datagrams without built-in delivery guarantees or ordering.
- DNS queries often use UDP, while applications may use TCP for larger responses or specific operational needs.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

04. IPv4 Addressing

An IPv4 address is 32 bits and is commonly written as four decimal octets. A subnet mask or CIDR prefix identifies the network portion. For example, /24 means 24 network bits and 8 host bits. Network and broadcast addresses are reserved in a conventional IPv4 subnet, while usable host counts depend on the subnet size and special-purpose ranges.

Key points

- An IPv4 address is 32 bits and is commonly written as four decimal octets.
- A subnet mask or CIDR prefix identifies the network portion.
- For example, /24 means 24 network bits and 8 host bits.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

05. Subnetting Practice

Subnetting divides a larger address block into smaller networks. A /27 prefix contains 32 total addresses, conventionally 30 usable host addresses. Subnet planning should consider current device counts, expected growth, routing boundaries, and broadcast-domain size. Incorrect masks or gateways can cause devices to be unable to reach peers or remote networks.

Key points

- Subnetting divides a larger address block into smaller networks.
- A /27 prefix contains 32 total addresses, conventionally 30 usable host addresses.
- Subnet planning should consider current device counts, expected growth, routing boundaries, and broadcast-domain size.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

06. IPv6 Overview

IPv6 uses 128-bit addresses written in hexadecimal groups. It offers a vastly larger address space and supports stateless address autoconfiguration and neighbor discovery. IPv6 does not use broadcast in the same way as IPv4; it relies on multicast and anycast. Security controls must cover IPv6 even when an organization primarily uses IPv4.

Key points

- IPv6 uses 128-bit addresses written in hexadecimal groups.
- It offers a vastly larger address space and supports stateless address autoconfiguration and neighbor discovery.
- IPv6 does not use broadcast in the same way as IPv4; it relies on multicast and anycast.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

07. Ethernet and Switching

Ethernet frames carry source and destination MAC addresses. Switches learn source MAC addresses and forward known unicast frames to the relevant port; unknown unicast and broadcast traffic may be flooded within the VLAN. VLANs create logical Layer 2 segments. Trunk links carry traffic for multiple VLANs using tagging such as IEEE 802.1Q.

Key points

- Ethernet frames carry source and destination MAC addresses.
- Switches learn source MAC addresses and forward known unicast frames to the relevant port; unknown unicast and broadcast traffic may be flooded within the VLAN.
- VLANs create logical Layer 2 segments.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

08. Routing and Default Gateways

Routers forward packets between IP networks using routing tables. A host sends off-subnet traffic to its default gateway. Routing decisions typically use the most specific matching route. A correct IP address alone is not enough: the subnet mask, gateway, route availability, and firewall policy must also be correct.

Key points

- Routers forward packets between IP networks using routing tables.
- A host sends off-subnet traffic to its default gateway.
- Routing decisions typically use the most specific matching route.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

09. DNS Resolution

DNS translates names to records such as A, AAAA, CNAME, and MX. A client may check a local cache before querying a recursive resolver. The resolver can consult authoritative servers and cache answers according to TTL values. DNS troubleshooting should compare name resolution with direct IP connectivity and check the configured resolver, suffix, and relevant records.

Key points

- DNS translates names to records such as A, AAAA, CNAME, and MX.
- A client may check a local cache before querying a recursive resolver.
- The resolver can consult authoritative servers and cache answers according to TTL values.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

10. DHCP Address Assignment

DHCP commonly uses a discover, offer, request, acknowledge exchange to lease network configuration. A lease may include an address, subnet mask, router, DNS servers, and lease duration. Address-pool exhaustion, incorrect VLAN placement, relay configuration, or rogue DHCP servers can disrupt clients. Static addresses must not conflict with the dynamic pool.

Key points

- DHCP commonly uses a discover, offer, request, acknowledge exchange to lease network configuration.
- A lease may include an address, subnet mask, router, DNS servers, and lease duration.
- Address-pool exhaustion, incorrect VLAN placement, relay configuration, or rogue DHCP servers can disrupt clients.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

11. Wireless LAN Basics

Wi-Fi networks use radio channels and access points to connect clients. Signal quality, interference, channel overlap, client density, and roaming behavior affect performance. A strong signal does not guarantee a fast connection because airtime is shared. Troubleshooting should distinguish authentication failure, association failure, DHCP failure, DNS failure, and poor application performance.

Key points

- Wi-Fi networks use radio channels and access points to connect clients.
- Signal quality, interference, channel overlap, client density, and roaming behavior affect performance.
- A strong signal does not guarantee a fast connection because airtime is shared.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

12. Wi-Fi Security

WPA2-AES and WPA3 are common Wi-Fi security options. Strong unique credentials, modern encryption, firmware updates, and disabling obsolete protocols reduce risk. Enterprise deployments may use 802.1X with an authentication server rather than a shared password. Guest access should be separated from sensitive internal systems.

Key points

- WPA2-AES and WPA3 are common Wi-Fi security options.
- Strong unique credentials, modern encryption, firmware updates, and disabling obsolete protocols reduce risk.
- Enterprise deployments may use 802.1X with an authentication server rather than a shared password.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

13. Firewalls and ACLs

A firewall enforces traffic policy using attributes such as source, destination, protocol, and port. Access control lists may be evaluated in order, so a broad earlier rule can shadow a more specific later rule. Least privilege means allowing only required traffic. Troubleshooting should inspect both directions of a flow and account for stateful behavior.

Key points

- A firewall enforces traffic policy using attributes such as source, destination, protocol, and port.
- Access control lists may be evaluated in order, so a broad earlier rule can shadow a more specific later rule.
- Least privilege means allowing only required traffic.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

14. Network Segmentation

Segmentation limits unnecessary communication between user devices, servers, management interfaces, and guest systems. VLANs provide Layer 2 separation, while routing and firewall rules enforce communication policy between segments. Segmentation can reduce lateral movement but does not replace endpoint security, identity controls, patching, or monitoring.

Key points

- Segmentation limits unnecessary communication between user devices, servers, management interfaces, and guest systems.
- VLANs provide Layer 2 separation, while routing and firewall rules enforce communication policy between segments.
- Segmentation can reduce lateral movement but does not replace endpoint security, identity controls, patching, or monitoring.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

15. VPN Concepts

A virtual private network creates a protected tunnel over an untrusted network. Remote-access VPNs connect individual users; site-to-site VPNs connect networks. Encryption protects traffic in transit, but weak credentials, compromised endpoints, overly broad routes, or poor authorization can still expose resources. Split tunneling changes which traffic travels through the tunnel.

Key points

- A virtual private network creates a protected tunnel over an untrusted network.
- Remote-access VPNs connect individual users; site-to-site VPNs connect networks.
- Encryption protects traffic in transit, but weak credentials, compromised endpoints, overly broad routes, or poor authorization can still expose resources.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

16. TLS and HTTPS

TLS protects data in transit through encryption, integrity checks, and server authentication using certificates. HTTPS is HTTP carried over TLS. Certificate validation includes checking the trusted chain, hostname, validity period, and relevant constraints. A certificate warning should not be bypassed casually because it may indicate misconfiguration or interception.

Key points

- TLS protects data in transit through encryption, integrity checks, and server authentication using certificates.
- HTTPS is HTTP carried over TLS.
- Certificate validation includes checking the trusted chain, hostname, validity period, and relevant constraints.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

17. Authentication and MFA

Authentication verifies identity; authorization determines what an authenticated identity may do. Multi-factor authentication combines different factor types, such as knowledge, possession, and inherence. MFA reduces risk from stolen passwords, although phishing-resistant methods provide stronger protection than one-time codes in many threat scenarios.

Key points

- Authentication verifies identity; authorization determines what an authenticated identity may do.
- Multi-factor authentication combines different factor types, such as knowledge, possession, and inherence.
- MFA reduces risk from stolen passwords, although phishing-resistant methods provide stronger protection than one-time codes in many threat scenarios.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

18. Least Privilege and RBAC

Least privilege grants only the permissions needed for a task and for the necessary duration. Role-based access control assigns permissions to roles and users to roles. Regular access reviews, separation of duties, and prompt removal of departed users reduce excess privilege. Shared accounts weaken accountability and should be avoided where practical.

Key points

- Least privilege grants only the permissions needed for a task and for the necessary duration.
- Role-based access control assigns permissions to roles and users to roles.
- Regular access reviews, separation of duties, and prompt removal of departed users reduce excess privilege.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

19. Endpoint Security

Endpoint controls include timely patching, anti-malware or endpoint detection, host firewalls, disk encryption, secure configuration, and least-privilege user accounts. Controls should be centrally monitored where possible. An endpoint tool is not a guarantee of safety; suspicious behavior still requires investigation and response.

Key points

- Endpoint controls include timely patching, anti-malware or endpoint detection, host firewalls, disk encryption, secure configuration, and least-privilege user accounts.
- Controls should be centrally monitored where possible.
- An endpoint tool is not a guarantee of safety; suspicious behavior still requires investigation and response.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

20. Patch and Vulnerability Management

Vulnerability management includes asset discovery, scanning, risk prioritization, remediation, and verification. Severity is one input; exposure, exploitability, asset importance, and compensating controls also matter. Patches should be tested appropriately and tracked to closure. Unsupported software requires migration or carefully documented risk mitigation.

Key points

- Vulnerability management includes asset discovery, scanning, risk prioritization, remediation, and verification.
- Severity is one input; exposure, exploitability, asset importance, and compensating controls also matter.
- Patches should be tested appropriately and tracked to closure.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

21. Phishing and Social Engineering

Phishing attempts to trick people into revealing information, opening malicious files, or approving fraudulent actions. Warning signs include unexpected urgency, mismatched sender details, unusual requests, and links to unfamiliar domains. Users should verify sensitive requests through a separate trusted channel and report suspicious messages rather than forwarding them broadly.

Key points

- Phishing attempts to trick people into revealing information, opening malicious files, or approving fraudulent actions.
- Warning signs include unexpected urgency, mismatched sender details, unusual requests, and links to unfamiliar domains.
- Users should verify sensitive requests through a separate trusted channel and report suspicious messages rather than forwarding them broadly.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

22. Malware and Ransomware

Malware includes malicious software such as trojans, spyware, and ransomware. Ransomware may encrypt files and attempt to disrupt recovery. Defense in depth includes patching, least privilege, email filtering, endpoint monitoring, network segmentation, and tested offline or immutable backups. Paying a ransom does not guarantee recovery or deletion of stolen data.

Key points

- Malware includes malicious software such as trojans, spyware, and ransomware.
- Ransomware may encrypt files and attempt to disrupt recovery.
- Defense in depth includes patching, least privilege, email filtering, endpoint monitoring, network segmentation, and tested offline or immutable backups.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

23. Logging and Monitoring

Useful logs record relevant events with reliable timestamps and sufficient context. Centralized collection helps correlate events across endpoints, servers, identity systems, and network devices. Monitoring should define alert thresholds, ownership, retention, access controls, and escalation procedures. Excessive noisy alerts can hide important events.

Key points

- Useful logs record relevant events with reliable timestamps and sufficient context.
- Centralized collection helps correlate events across endpoints, servers, identity systems, and network devices.
- Monitoring should define alert thresholds, ownership, retention, access controls, and escalation procedures.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

24. Incident Response

Incident response commonly includes preparation, detection and analysis, containment, eradication, recovery, and lessons learned. Preserve evidence and record actions. Containment choices should consider business impact and risk of continued compromise. Communication should follow the organization's incident plan, including privacy, legal, and regulatory requirements where applicable.

Key points

- Incident response commonly includes preparation, detection and analysis, containment, eradication, recovery, and lessons learned.
- Preserve evidence and record actions.
- Containment choices should consider business impact and risk of continued compromise.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

25. Backups and Recovery

A backup strategy should define recovery point objective (RPO) and recovery time objective (RTO). Backups should be protected from ordinary administrator compromise, monitored for failures, and tested through restoration exercises. A successful backup job does not prove that the data can be restored; recovery tests validate usability and completeness.

Key points

- A backup strategy should define recovery point objective (RPO) and recovery time objective (RTO).
- Backups should be protected from ordinary administrator compromise, monitored for failures, and tested through restoration exercises.
- A successful backup job does not prove that the data can be restored; recovery tests validate usability and completeness.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

26. Availability and Redundancy

Availability depends on resilient components, capacity, monitoring, and recovery processes. Redundant links or devices help only when failure modes are independent and failover is tested. High availability does not eliminate the need for backups, security controls, or disaster recovery planning. Single points of failure should be identified and prioritized.

Key points

- Availability depends on resilient components, capacity, monitoring, and recovery processes.
- Redundant links or devices help only when failure modes are independent and failover is tested.
- High availability does not eliminate the need for backups, security controls, or disaster recovery planning.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

27. Network Performance

Latency measures delay, jitter describes variation in delay, throughput measures achieved data transfer, and packet loss measures packets that fail to arrive. High latency can hurt interactive applications even when throughput is high. Troubleshooting should compare wired and wireless paths, local and remote destinations, and measurements over time.

Key points

- Latency measures delay, jitter describes variation in delay, throughput measures achieved data transfer, and packet loss measures packets that fail to arrive.
- High latency can hurt interactive applications even when throughput is high.
- Troubleshooting should compare wired and wireless paths, local and remote destinations, and measurements over time.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

28. Packet Capture Basics

Packet captures provide evidence about protocols, timing, retransmissions, handshakes, and failures. Captures may contain sensitive information and should be collected with authorization, minimized, stored securely, and deleted according to policy. A capture is a view of observed traffic, not necessarily a complete representation of all network activity.

Key points

- Packet captures provide evidence about protocols, timing, retransmissions, handshakes, and failures.
- Captures may contain sensitive information and should be collected with authorization, minimized, stored securely, and deleted according to policy.
- A capture is a view of observed traffic, not necessarily a complete representation of all network activity.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

29. Secure Remote Administration

Administrative access should use approved secure protocols, strong authentication, restricted source networks, and dedicated accounts. Management interfaces should not be exposed broadly to the internet. Administrative actions should be logged, and emergency access should be controlled and reviewed. Default credentials and unnecessary management services should be removed.

Key points

- Administrative access should use approved secure protocols, strong authentication, restricted source networks, and dedicated accounts.
- Management interfaces should not be exposed broadly to the internet.
- Administrative actions should be logged, and emergency access should be controlled and reviewed.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

30. Cloud Shared Responsibility

Cloud security responsibilities are shared between the provider and customer, with exact duties depending on the service model. Providers may secure physical infrastructure while customers remain responsible for identity, data classification, access policies, and configuration. Misconfigured storage, excessive permissions, and exposed secrets are common preventable risks.

Key points

- Cloud security responsibilities are shared between the provider and customer, with exact duties depending on the service model.
- Providers may secure physical infrastructure while customers remain responsible for identity, data classification, access policies, and configuration.
- Misconfigured storage, excessive permissions, and exposed secrets are common preventable risks.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

31. API Security

APIs should authenticate callers, authorize each requested resource, validate input, limit abusive request rates, and avoid exposing secrets in URLs or logs. Authentication alone does not prevent broken object-level authorization. Versioning, monitoring, schema validation, and safe error handling improve reliability and security.

Key points

- APIs should authenticate callers, authorize each requested resource, validate input, limit abusive request rates, and avoid exposing secrets in URLs or logs.
- Authentication alone does not prevent broken object-level authorization.
- Versioning, monitoring, schema validation, and safe error handling improve reliability and security.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

32. Data Privacy and Handling

Data should be classified by sensitivity and handled according to retention, access, and disclosure requirements. Collect only what is needed, restrict access, encrypt sensitive data appropriately, and remove it securely when no longer required. Logs and test datasets can themselves contain personal or confidential information.

Key points

- Data should be classified by sensitivity and handled according to retention, access, and disclosure requirements.
- Collect only what is needed, restrict access, encrypt sensitive data appropriately, and remove it securely when no longer required.
- Logs and test datasets can themselves contain personal or confidential information.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

33. Secure Configuration

Secure baselines disable unnecessary services, remove default credentials, limit administrative access, and apply appropriate protocol settings. Configuration drift can reintroduce risk after an initial hardening effort. Automated checks and documented change control help maintain a known-good state.

Key points

- Secure baselines disable unnecessary services, remove default credentials, limit administrative access, and apply appropriate protocol settings.
- Configuration drift can reintroduce risk after an initial hardening effort.
- Automated checks and documented change control help maintain a known-good state.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

34. Zero Trust Principles

Zero trust avoids granting trust solely because a request originates inside a network. Access decisions should consider identity, device posture, resource sensitivity, and context, with ongoing verification where appropriate. It is an architecture approach rather than a single product, and it still depends on sound monitoring and operational practices.

Key points

- Zero trust avoids granting trust solely because a request originates inside a network.
- Access decisions should consider identity, device posture, resource sensitivity, and context, with ongoing verification where appropriate.
- It is an architecture approach rather than a single product, and it still depends on sound monitoring and operational practices.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

35. Common Troubleshooting Workflow

A disciplined workflow defines the symptom, scope, timeline, and recent changes before testing hypotheses. Check physical/link status, addressing, gateway reachability, DNS, routing, policy, and application behavior in a structured order. Change one variable at a time where possible and record the result so that fixes are reproducible.

Key points

- A disciplined workflow defines the symptom, scope, timeline, and recent changes before testing hypotheses.
- Check physical/link status, addressing, gateway reachability, DNS, routing, policy, and application behavior in a structured order.
- Change one variable at a time where possible and record the result so that fixes are reproducible.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

36. Outage Case Study: DNS

If users cannot open a service by name but can reach its IP address, DNS is a strong hypothesis, though application behavior and virtual hosting may complicate the test. Compare results from the configured resolver and a known authorized resolver, inspect record values and TTL, and verify that the hostname points to the expected service.

Key points

- If users cannot open a service by name but can reach its IP address, DNS is a strong hypothesis, though application behavior and virtual hosting may complicate the test.
- Compare results from the configured resolver and a known authorized resolver, inspect record values and TTL, and verify that the hostname points to the expected service.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

37. Outage Case Study: DHCP

If new clients fail while existing clients continue working, inspect the DHCP scope, lease availability, VLAN placement, relay configuration, and server health. A client with an automatic private address may not have received a usable lease. Avoid assigning random static addresses because this can create conflicts and obscure the underlying fault.

Key points

- If new clients fail while existing clients continue working, inspect the DHCP scope, lease availability, VLAN placement, relay configuration, and server health.
- A client with an automatic private address may not have received a usable lease.
- Avoid assigning random static addresses because this can create conflicts and obscure the underlying fault.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

38. Outage Case Study: Firewall

If a service is reachable from one network but not another, compare source and destination addresses, ports, routes, and firewall logs. Confirm whether the service is listening and whether return traffic is permitted. Avoid solving the issue by broadly disabling the firewall; make a narrow, reviewed rule change and verify the result.

Key points

- If a service is reachable from one network but not another, compare source and destination addresses, ports, routes, and firewall logs.
- Confirm whether the service is listening and whether return traffic is permitted.
- Avoid solving the issue by broadly disabling the firewall; make a narrow, reviewed rule change and verify the result.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

39. Evidence-Based RAG Answers

A retrieval-augmented generation system retrieves relevant document chunks and uses them as context for a language model. Reliable answers should be supported by retrieved evidence, distinguish evidence from inference, and cite source documents or page references when available. Retrieval relevance alone does not prove that a generated claim is faithful to the sources.

Key points

- A retrieval-augmented generation system retrieves relevant document chunks and uses them as context for a language model.
- Reliable answers should be supported by retrieved evidence, distinguish evidence from inference, and cite source documents or page references when available.
- Retrieval relevance alone does not prove that a generated claim is faithful to the sources.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.

40. RAG Evaluation and Abstention

Faithfulness asks whether answer claims are supported by the supplied context. Answer relevance checks whether the response addresses the question. Context relevance considers whether retrieved material is useful for the question. When evidence is insufficient, a system should state the limitation or abstain instead of inventing unsupported details. Automated metrics are useful signals, not infallible ground truth.

Key points

- Faithfulness asks whether answer claims are supported by the supplied context.
- Answer relevance checks whether the response addresses the question.
- Context relevance considers whether retrieved material is useful for the question.

Evaluation note: A supported answer should stay within these statements, avoid adding ungrounded specifics, and identify when the source does not provide enough information.